

Qatar-1b

R-0610

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230429 · created 2026-07-25T07:24:54+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460062.4848 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.056
Transit depth (Rp/Rs)^2	3.09 +/- 1.96 [%]
Orbital Inclination (inc)	88.71 +/- 2.85
Airmass coefficient 1 (a1)	0.0062 +/- 0.0029
Airmass coefficient 2 (a2)	2.98 +/- 1.49
Scatter in the residuals of the lightcurve fit is	52.65 %
Best Comparison Star	#2 - [541, 468]
Optimal Aperture	1.46
Optimal Annulus	7.25
Transit Duration (day)	0.0053 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.176 $\pm$ 0.056 vs 0.1463 (deviation 0.53 σ)
Duration fit vs published	0.0053 d vs 0.0692 d (deviation 18.25 σ)
Mid-time O-C	-2050.9 min
Depth SNR	3.1
Residual scatter	52.65 %

Light curve

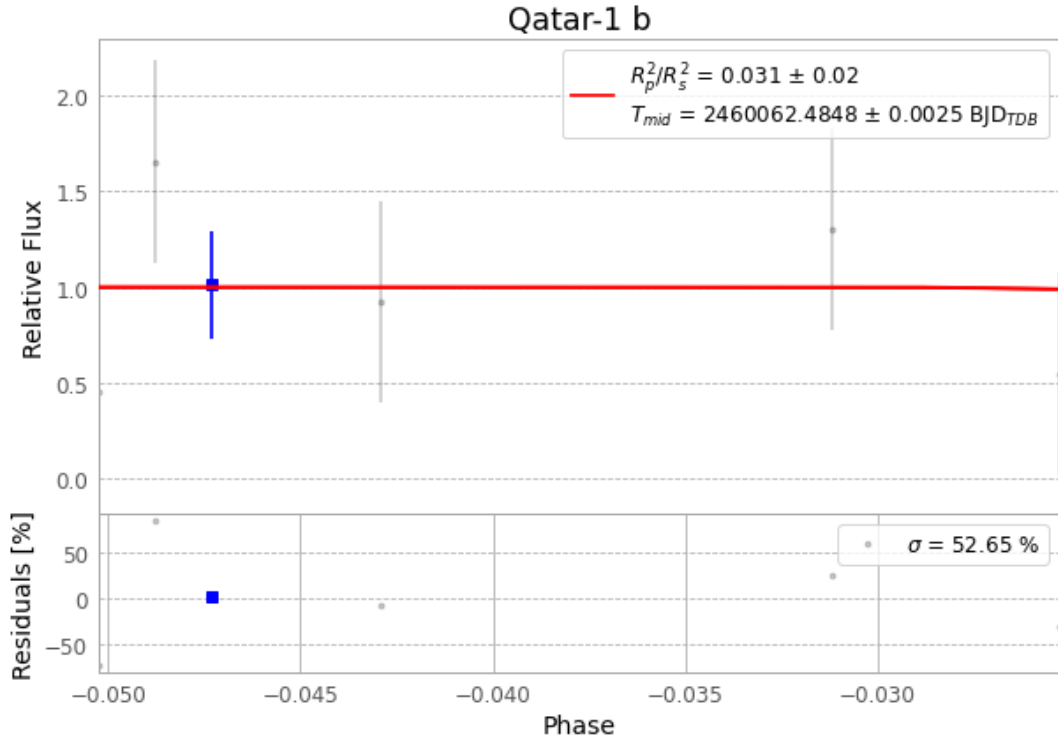


Figure 1 — FinalLightCurve_Qatar-1 b_2023-04-29.png (SHA-256 e90e93a676bd01f6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [469, 426], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 12a00c7586407d79...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

74 FITS frames (49 MB), Qatar-1230429080015.FITS → Qatar-1230429113915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230429.log (4d3efebd52b6...), FinalLightCurve_Qatar-1 b_2023-04-29.png (e90e93a676bd...), FinalLightCurve_Qatar-1 b_2023-04-29.csv (c3e711981381...)

Compute resources

Wall time 80.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 07:24Z.